

Swin-SegFormer: Semantic Segmentation of Aerial Drone Imagery

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TL;DR — Swin Transformer encoder + SegFormer MLP decoder. Trained on Aerialscapes (3,269 drone images, 12 classes) with no pretrained weights. **Best: 65.69% val mIoU — 51.08% test mIoU.**

Introduction

Semantic segmentation assigns a class to every pixel enabling drones to detect landing zones, supporting urban planning by detecting roads, buildings, etc.

Key challenges:

- Small dataset: only 3,269 images, no pretraining.
- Class imbalance: vegetation + road + sky > 70% of pixels. drone + boat < 0.5%.
- Altitude variability: imagery from 5 to 50 meters.

Dataset: Aerialscapes



Figure 1. Aerialscapes drone imagery at varying altitudes.

Train	2,096
Validation	525
Test (held-out)	648

12 classes: bg, person, bike, car, drone, boat, animal, obstacle, construction, vegetation, road, sky

Training

Loss	Weighted CE + Dice (no background)
Optimiser	AdamW
LR enc./dec.	6×10^{-5} / 6×10^{-4}
Regularisation	Dropout + DropPath 0.1
Epochs / Best	100 / 95

Augmentation: random flip, rotation $\pm 15^\circ$, scale crop (0.75-1.25), color jitter.

Background excluded from Dice loss: its pixel count is dominant and will mask poor performance on other classes.

Class weights are computed on training samples using inverse-class frequency, amplifying gradient signal for rare classes.

Architecture

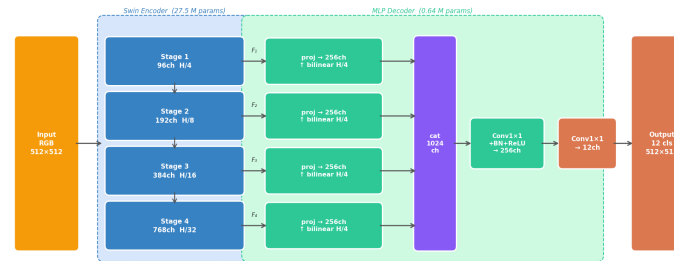


Figure 2. Swin-SegFormer architecture. Total: 28.2 M parameters.

Swin Transformer Encoder 27.5 M params

A **hierarchical vision transformer** with 4 stages, each halving resolution and doubling channels. Attention is confined to local **$M \times M$ windows** ($M = 8$), every other block **shifts the grid by $M/2$** to propagate information across boundaries.

Stage	Channels	Resolution	Heads	Blocks
1	96	$H/4 \times W/4$	3	2
2	192	$H/8 \times W/8$	6	2
3	384	$H/16 \times W/16$	12	6
4	768	$H/32 \times W/32$	24	2

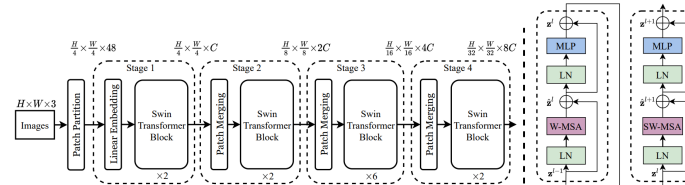


Figure 3. Swin Transformer architecture and successive W-MSA / SW-MSA blocks. From Liu et al., ICCV 2021.

MLP Decoder 0.64 M params

Inspired by **SegFormer**, the decoder is lightweight.

1. Each feature map linearly projected to **$D = 256$ ch**
2. All 4 maps upsampled to **$H/4 \times W/4$**
3. Concatenated $\rightarrow$ **1,024 ch** then fused via Conv1x1 + BN + ReLU
4. Final Conv1x1 $\rightarrow$ **12 class logits**, upsampled to 512x512

Training dynamics

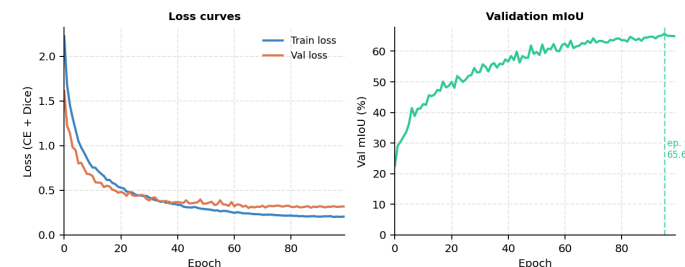


Figure 4. Loss curves and val mIoU over 100 epochs. Best: 65.69% at epoch 95.

Results

Qualitative

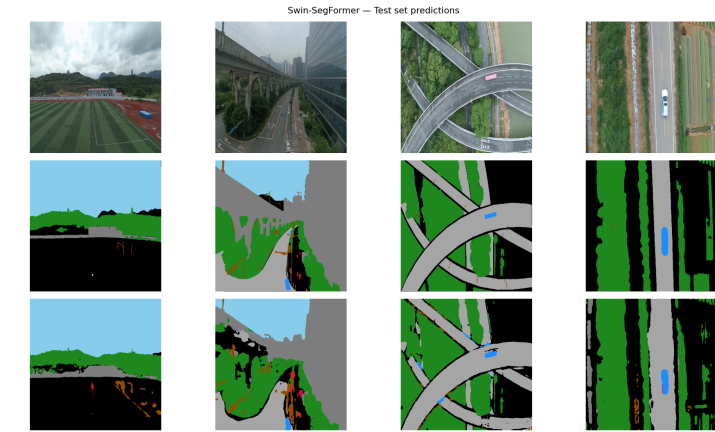


Figure 5. Test set results. Top: input. Middle: ground truth. Bottom: prediction.

Per-class IoU

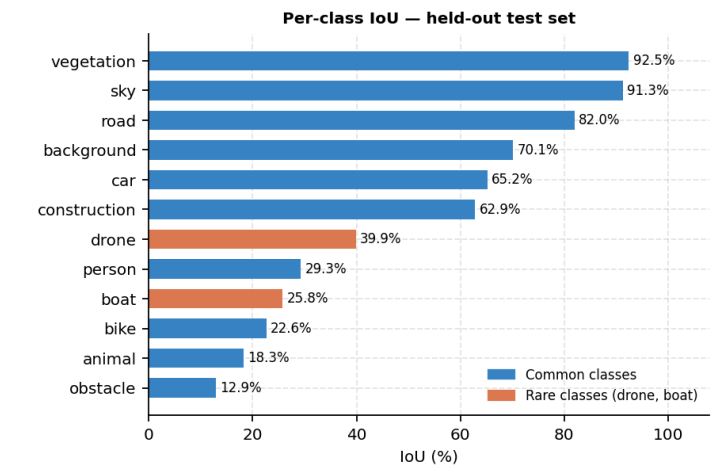


Figure 6. Per-class IoU on the held-out test set. Rare classes (drone, boat) in orange.

Model	Params	mIoU
Swin-SegFormer — val	28.2 M	65.69%
Swin-SegFormer — test	28.2 M	51.08%

Conclusion

Swin-SegFormer trained from scratch achieves **65.69% val mIoU** on Aerialscapes. Viable without pretraining. Class weighting and background excluded Dice were probably critical for rare class performance.

Future work: Pretraining, Swin-Small encoder, aspect-ratio preserving input (640x384), retrain on full train+val set.